

SMART INDIA HACKATHON 2026



FIG.1 SYSTEM IDENTITY

RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

SIH26002

PROBLEM STATEMENT ID
MDoNER · Smart Automation · Software

FIG.2 SUBMISSION RECORD

PS TITLE	AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)
THEME	Smart Automation
PS CATEGORY	Software
TEAM ID	17 (internal group no.)
TEAM NAME	NER-AI LOGISTICS

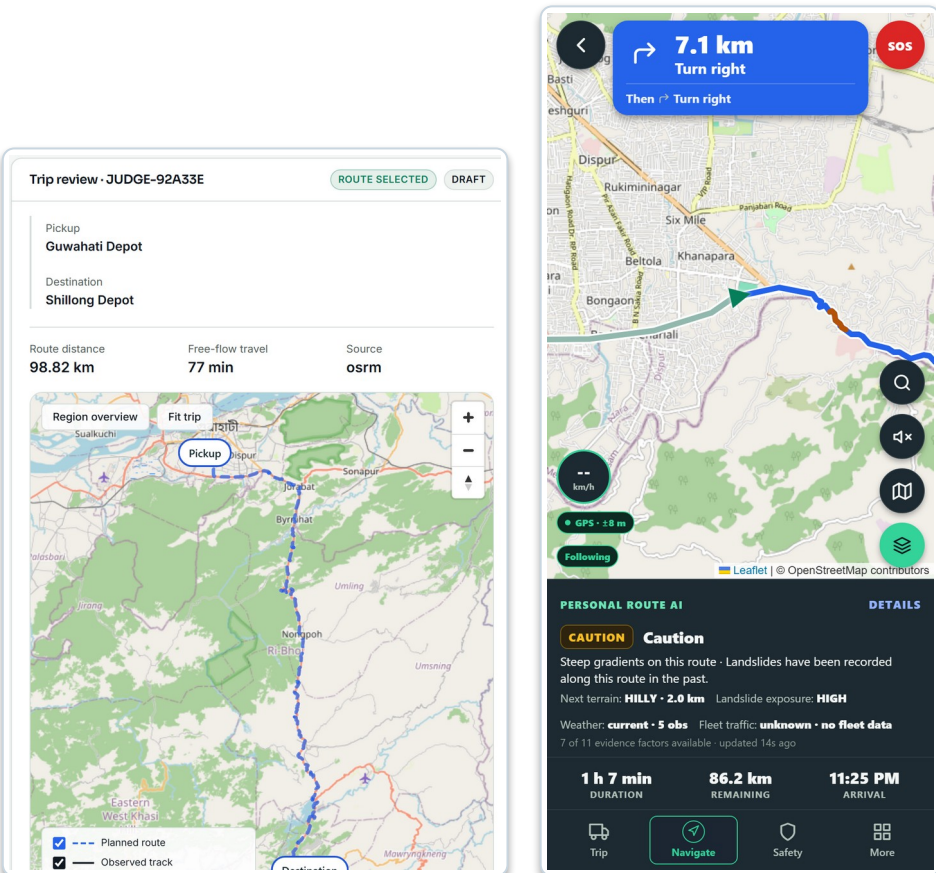
FIG.3 DESIGN PREMISE

SHORTEST IS NOT ALWAYS USABLE.

“Can this essential-supply truck reliably use this corridor now?”

End-to-end working prototype · physical-phone validated · live demo corridor Guwahati → Shillong

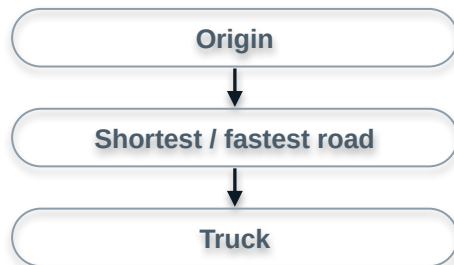
FIG.4 AS BUILT – MANAGER + DRIVER



REAL SCREENS / NO MOCK-UPS

Not the shortest road. The road that is operationally usable now.

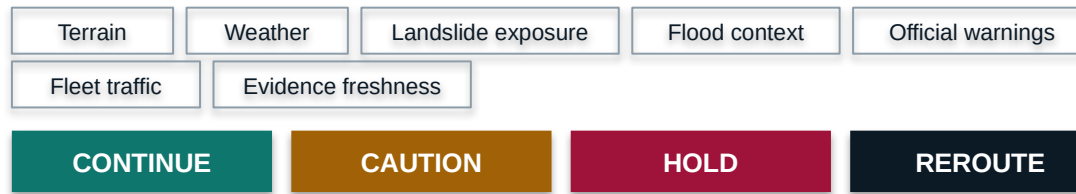
FIG.1 BASELINE



No governance. No answer when data is missing.

FIG.2 RASTA AI — EVIDENCE PATH

Origin → real OSRM route → route-specific evidence → policy → decision



Manager governance → driver navigation, 23 languages, offline package

FIG.3 THE INVARIANT

UNKNOWN ≠ SAFE

Missing or stale evidence is marked UNKNOWN, never counted as safe, and the route goes to an authorised reviewer.

FIG.4 PROBLEM CONTEXT

NER corridors close without notice: monsoon rain, landslides, floods, official warnings, single-lane hill roads, no phone signal. A shortest-path router still sends the truck.

FIG.5 MANAGER · CHECK CONDITIONS

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN

0.5 KM STEEP

49-1436 m · 1975 m climb · steepest 10.3%
flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY

HIGH EXPOSURE

22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog · inventory 2007-2017 — aged, recent years not covered

RIVER LEVELS

NORMAL

Discharge at 5 river cells along the corridor, highest 1.0x its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS

NONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umiling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC

UNKNOWN

EVIDENCE / FRESHNESS / UNKNOWN

FIG.1 VERIFIED DATA SOURCES

WEATHER	TERRAIN	WARNINGS	FLOOD	LANDSLIDE HISTORY	TRAFFIC
Open-Meteo · MET Norway	Copernicus DEM · OpenTopoData	NDMA SACHET (CAP)	GloFAS	NASA GLC 2007–17	fleet GPS fixes

FIG.2 DECISION PATH

ROUTE-SPECIFIC EVIDENCE · FastAPI backend

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, alternatives, turn steps

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE

CAUTION

HOLD

REROUTE

MANAGER REVIEW

DRIVER NAVIGATION

ROUTE-AHEAD 60 s

CONDITIONS CHANGE?

REASSESS

FIG.3 IMPLEMENTATION

CLIENTS Manager console — React + TypeScript, manager accounts only · Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package

STACK FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

EVIDENCE Open-Meteo / MET Norway · NDMA SACHET · GloFAS · NASA historical landslides · OpenTopoData / Copernicus DEM
Re-scored every 60 s along the road ahead; a reroute is proposed to the manager, never applied silently.

FIG.4 DECISION AUTHORITY

DETERMINISTIC SAFETY ENGINE

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision in the driver's language — never decides

EXPERIMENTAL ML

research channel only; held outside routing by the safety gate

GOVERNANCE CHAIN

Verified evidence



Deterministic safety policy



Manager governance



Driver action

Not a concept. A working end-to-end prototype.

FIG.1 CERTIFIED LIFECYCLE



FIG.2 AS BUILT

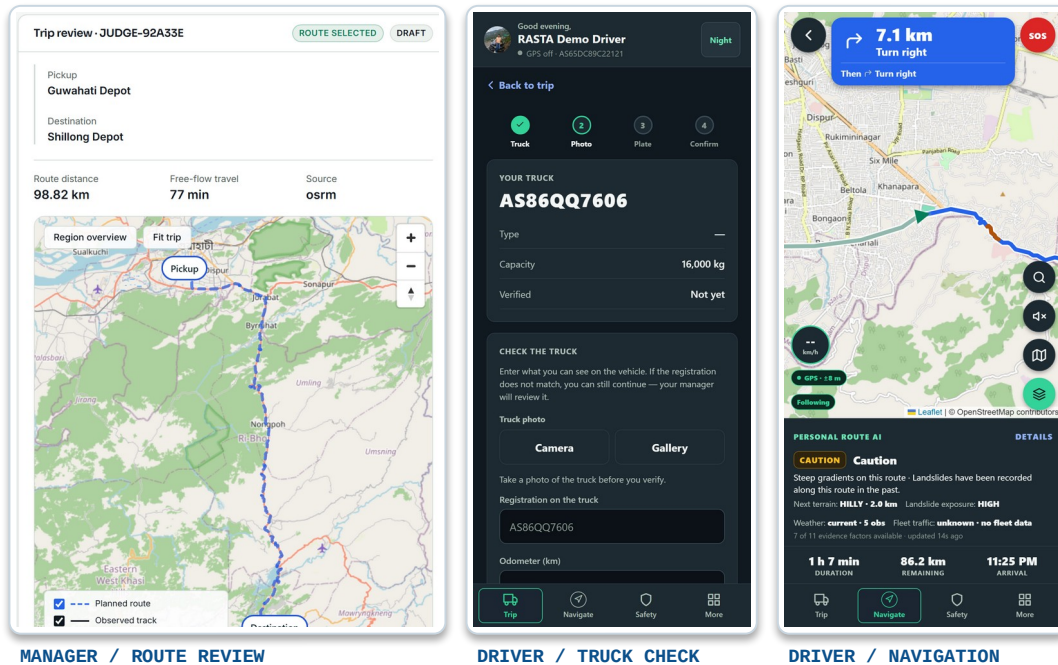


FIG.3 EVIDENCE OF EXECUTION

The whole lifecycle above was run end to end on a physical Android phone against the hosted backend, on the real road.

12/12 physical judge-flow steps

9/9 physical role-flow steps

1138+ backend · **621+** driver · **170+** manager tests

PHYSICAL ANDROID · CERTIFIED

CANONICAL DEMO: GUWAHATI → SHILLONG / ≈ 98.8 KM

KNOWN LIMITS → MITIGATION Render cold start → warm /health before the demo · provider rate limits → cached evidence + explicit UNKNOWN · background push needs Firebase config → in-app alerts already work · experimental ML → outside routing until the gate passes

RASTA AI does not ask only “Which road is shortest?” It asks “Can this truck reliably use this corridor now?”

FIG.1 BENEFICIARIES

FLEET MANAGERS

Route usability and uncertainty visible before dispatch; auditable decisions.

DRIVERS

Safer navigation, danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

More resilient access to food, medicine, fuel and relief supplies.

GOVERNMENT / OPERATIONS

Auditable route evidence and disruption visibility per corridor.

FIG.2 MEASURED TODAY

23 selectable languages, explicit fallback status

11 route factors in every decision

60 s route-ahead monitor while driving

12/12 judge-flow steps certified on a physical phone

100 % of reroutes reviewed by a human

0 unknown factors counted as safe

FIG.3 GOVERNANCE — AI HAS NO UNCONTROLLED AUTHORITY

FOOD · MEDICINE · FUEL · RELIEF SUPPLIES — essential logistics for hill communities, decided on evidence

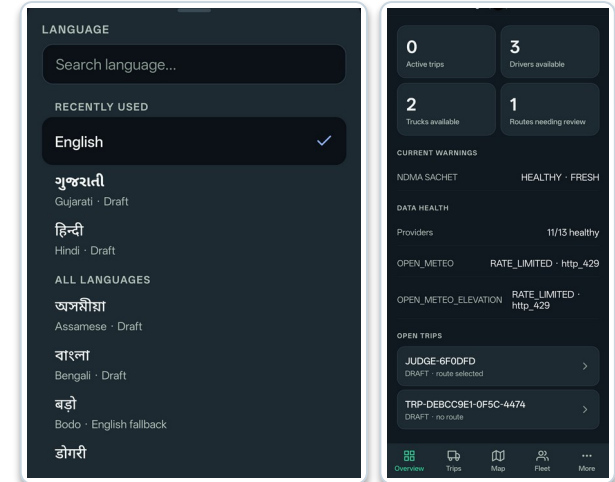
Verified evidence

Deterministic safety policy

Manager governance

Driver action

FIG.4 AS BUILT — APK 1.0.18



23 LANGUAGES

MANAGER / MOBILE

Real screens, APK 1.0.18. One login, server-decided role: a manager gets a mobile fleet view, a driver gets navigation. Each language shows its status — Verified, Draft or English fallback — so nobody is misled.

FIG.1 EVIDENCE SOURCES BEHIND THE ROUTE DECISION

ROUTING / PLACES

OSRM · OpenStreetMap · Nominatim

real road geometry, alternatives and turn steps; place search on the corridor

OFFICIAL / FLOOD

NDMA SACHET (CAP) · Copernicus GloFAS

official warnings polled with freshness; river discharge against its 30-day mean

AI ASSISTANCE

Google Gemini

explains the decision in the driver's language — never safety authority

WEATHER / TERRAIN

Open-Meteo · MET Norway · Copernicus DEM · OpenTopoData

forecast sampled along the road; climb, gradient and steep segments (GLO-90)

LANDSLIDE HISTORY

NASA Global Landslide Catalog

historical exposure; inventory 2007–2017, ≤5 km location-accuracy filter

PROBLEM STATEMENT

MDoNER · SIH26002

AI-based smart logistics and accessibility intelligence for the NER

FIG.2 EXPERIMENTAL LANDSLIDE RESEARCH

95 %

RECALL

on the NER geographic holdout — 154 catalogued events, no NER row used in training

SAFETY GATE

NOT DEPLOYED**FPR 50 %** too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. Strong recall was not enough. Held-out testing exposed the false-positive rate, so the gate keeps the model out of routing.

Precision 0.32 · F1 0.48 · PR-AUC 0.55 · ROC-AUC 0.83 · Brier 0.12 · calibrated logistic regression on NASA GLC + ERA5-Land rain + DEM slope · docs/HAZARD_ERROR_ANALYSIS.md